Mushfiqa Shaikh

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Professional Summary

Final-year B.Tech Computer Science Engineering (IoT) student with hands-on experience in embedded systems, IoT development, sensor integration, microcontrollers, and wireless communication. Skilled in C++, Python, Arduino, ESP32, embedded firmware development, hardware interfacing, and Firebase. Experienced in designing, prototyping, testing, and debugging sensor-based IoT systems. Interested in embedded systems, robotics, IoT hardware, and real-time connected applications.

Education

Parul Institute Of Engineering And Technology
B.Tech Computer Science Engineering (IoT) | 2023–2027
CGPA: 8.05/10

Technical Skills

Programming: C, C++, Python, JavaScript
Embedded Systems: ESP32, Arduino, Raspberry Pi, Microcontrollers, Firmware Development, Sensor Integration, Hardware Interfacing, Circuit Prototyping, Sensor Calibration, Serial Communication
IoT: IoT Architecture, Real-Time IoT Systems, Wireless Communication, Firebase Realtime Database, MQTT
Communication Protocols: UART, I2C, SPI, HTTP/HTTPS
Cloud & IoT Platforms: Firebase, ThingSpeak
Tools: Arduino IDE, Git, GitHub
Web Technologies: HTML5, CSS3, JavaScript, React.js

Projects

Smart Water Quality Monitoring System In Real Time — GitHub Repo

- Engineering a scalable IoT platform for real-time university water quality monitoring using ESP32 and Firebase concepts.
- Measures pH, TDS, turbidity, temperature, EC, and water level while calculating Water Quality Index (WQI).
- **Developing a cloud-based dashboard for real-time visualization, intelligent alerts, and historical analytics.**
- Technologies: ESP32, Sensor Integration, Arduino IDE, C++, Firebase, Bootstrap, CSS, HTML, Javascript, Chart.js, Git/GitHub.

Hand Gesture to Text Conversion System — GitHub Repo

- Developed an embedded assistive communication prototype using ESP32 for converting hand gestures into text.
- Integrated MPU6050 motion sensing with the microcontroller for gesture data acquisition and recognition.
- Developed embedded C++ firmware for sensor interfacing, calibration, and gesture classification.
- Tested sensor responses and refined gesture recognition logic to improve system reliability.

Technologies: ESP32, MPU6050, C++, Arduino IDE, Embedded Systems, Sensor Integration

Hackathons & Activities

- Participated and led teams in multiple competitive hackathons focused on IoT, healthcare technology, embedded systems, and web development.
- Organized and managed technical events, and workshops as a Core Committee Member of the IoT Club.

Areas of Interest

IoT, Embedded Systems, Robotics, Software Engineering, Cloud Computing